

Range function

You are given an array A of N integer elements. You are also given Q queries, where each query is one of the following types:

- $1\ i\ val$: Update value of element at i -th index to val i.e. $A[i] = val$

- $2\ L\ R$: Find the value of function $\sum_{i=L}^R \sum_{j=i}^R F(i, j)$ where $F(i, j)$

represents the sum of elements of array A in index range L to

R .

Task

Assumptions

- $N = 5$
- $Q = 2$
- $A = [2, 1, 4, 3, 1]$
- $query = [(1, 2, 2), (2, 1, 3)]$

Approach

- First query is of Type 1, Update $A[2] = 2$. Now, array A becomes $[2, 2, 4, 3, 1]$.
- Second query of of Type 2 with $L = 1, R = 3$.
 - Value of function is equal to $F(1, 1) + F(1, 2) + F(1, 3) + F(2, 2) + F(2, 3) + F(3, 3) = 2 + 4 + 8 + 2 + 6 + 4 = 26$.

Thus, the answer is 26.

Question 1

Max. score: 30.00

1

First subsequence

2

You are given the following:

- String *A* consists of lowercase English letters
- String *B* consists of lowercase English letters

You have to check whether string *A* contains string *B* as a subsequence or not. You can change at most 1 character in string *B* (except the first character).

Task

Determine the first index of occurrence of string *B* in string *A*. If not found, output -1.

Notes

- A string *b* is said to be a subsequence of string *a*, if *b* can be obtained from *a* by replacing at most 1 character without changing the ordering of the remaining characters.
- 1 based indexing is used.

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25

26

0 / 2 Completed

Notes

- A string b is said to be a subsequence of string a , if b can be obtained from a by replacing at most 1 character without changing the ordering of the remaining characters.
- 1 based indexing is used.

Example

Assumptions

- $A = daabe$
- $B = abe$

Approach

The subsequences of string A that can match with string B by changing at most 1 character are:

- $A[2], A[3], A[5] : aae$ (requires a change of character $B[2]$ from 'b' to 'a').
- $A[2], A[4], A[5] : abe$ (requires no change).
- $A[3], A[4], A[5] : abe$ (requires no change).

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Therefore, you can find the first occurrence of string B from the 2nd index in A . Hence, the answer is 2.

Function description

Complete the *firstOccurence* function provided in the editor. This function takes the following 2 parameters and returns the answer:

- A : Represents the first string
- B : Represents the second string

Input format

Note: This is the input format that you must use to provide custom input (available above the **Compile and Test** button).

- The first line contains an integer T denoting the number of test cases. T also denotes the number of times you have to run the *firstOccurence* function on a different set of inputs.
- For each test case:
 - The first line contains a string A consists of lowercase English letters.
 - The second line contains a string B of consists of lowercase

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cases. T also denotes the number of times you have to run the *firstOccurrence* function on a different set of inputs.

- For each test case:
 - The first line contains a string A consists of lowercase English letters.
 - The second line contains a string B of consists of lowercase English letters.

Output format

For each test case, print the answer in a new line.

Constraints

$$1 \leq T \leq 10$$

$$1 \leq |A| \leq 2000$$

$$1 \leq |B| \leq 2000$$

Code snippets (also called starter code/boilerplate code)

This question has code snippets for C, CPP, Java, and Python.

Sample input



Sample output



2

3

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Code snippets (also called starter code/boilerplate code)

This question has code snippets for C, CPP, Java, and Python.

Sample input

```
2
abcbc
cba
lhs
rhs
```

Sample output

```
3
-1
```

Explanation

The first line represents the number of test cases, $T = 2$.

For test case 1

Given

- $A = abcbc$
- $B = cba$

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The first line represents the number of test cases, $T = 2$.

For test case 1*Given*

- $A = abcbc$
- $B = cba$

Approach

The subsequences of string A that can match with string B by changing at most 1 character are:

- $A[3], A[4], A[5] : cbc$ (requires a change of character $B[3]$ from 'a' to 'c').

Therefore, you can find the first occurrence of string B from the 3rd index in A .

Hence, the answer is 3.

For test case 2*Given*

- $A = lhs$
- $B = rhs$

Approach

- You can not change the first character of string B .
- Therefore, you can not find the occurrence of string B in A .

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- $A = lhs$
- $B = rhs$

Approach

- You can not change the first character of string B .
- Therefore, you can not find the occurrence of string B in A .

Hence, the answer is -1.

Note:

Your code must be able to print the sample output from the provided sample input. However, your code is run against multiple hidden test cases. Therefore, your code must pass these hidden test cases to solve the problem statement.

Limits

Time Limit: 1.0 sec(s) for each input file

Memory Limit: 256 MB

Source Limit: 1024 KB

Scoring

Score is assigned if any testcase passes

Allowed Languages

Bash, C, C++14, C++17, Clojure, C#, D, Erlang, F#, Go, Groovy, Haskell, Java 8, Java 14, Java 17, JavaScript(Node.js), Julia, Kotlin, Lisp (SBCL), Lua, Objective-C, OCaml, Octave, Pascal, Perl, PHP, Python, Python 3, Python 3.8, R(RScript), Racket, Ruby, Rust, Scala, Swift, TypeScript, Visual Basic

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00:58:51

902466566600@googl...

C++17 (g++ 10.3.0)



Auto-complete ready!

```
1  #include<bits/stdc++.h>
2  using namespace std;
3
4  ✓ int firstOccurence (string A, string B) {
5      // Write your code here
6
7  }
8
9  ✓ int main() {
10
11      ios::sync_with_stdio(0);
12      cin.tie(0);
13      int T;
14      cin >> T;
15      for(int t_i = 0; t_i < T; t_i++)
16  ✓ {
17          string A;          I
18          cin >> A;
19          string B;
20          cin >> B;
21
22          int out_;
23          out_ = firstOccurence(A, B);
24          cout << out_;
25          cout << "\n";
26      }
```

[View test cases](#) ^



29°C
Mostly cloudy



ENG
IN



Question 2

Max. score: 30.00

OR XOR

You are given the following:

- Two integers N and K
- An array A of size N

Let sequence $B(B_1, B_2, \dots, B_{2K})$ of size $2K$ is a subsequence of array A . Let's define $F(B) = (B_1 | B_2 | \dots | B_K) \oplus (B_{K+1} | B_{K+2} | \dots | B_{2K})$.

Where $|$ denotes bitwise or operation and $\oplus$ denotes bitwise XOR operation.

Task

Determine the maximum value of $F(B)$ for all possible sequences B .

Note: A sequence X is a subsequence of sequence Y if X can be obtained from Y by deleting several (possibly, zero or all) elements without changing the order of the remaining elements.

Example

Assumptions

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Example

Assumptions

- $N = 3$
- $K = 1$
- $A = [2, 4, 5]$

Approach There are 3 subsequences of length 2 which are as follows:

- $B = [2, 4], F(B) = 2 \oplus 4 = 6$
- $B = [2, 5], F(B) = 2 \oplus 5 = 7$
- $B = [4, 5], F(B) = 4 \oplus 5 = 1$

The maximum value of $F(B)$ is 7, thus the answer is 7.

Function description

Complete the `solve` function provided in the editor. This function takes the following 3 parameters and returns a number:

- N : Represents the integer N
- K : Represents the integer K
- A : Represents an integer array of size N denoting A

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- K : Represents the integer K
- A : Represents an integer array of size N denoting A

Input format

Note: This is the input format that you must use to provide custom input (available above the **Compile and Test** button).

- The first line contains a single integer T that denotes the number of test cases. T also denotes the number of times you have to run the *solve* function on a different set of inputs.
- For each test case:
 - The first line contains an integer N .
 - The second line contains an integer K .
 - The third line contains N space-separated integers denoting the array A .

Output format

For each test case in a new line, print the answer representing the maximum value of $F(B)$ for all possible sequences B .

Constraints

$$1 \leq T \leq 100, 1 \leq N \leq 1001, K \leq 10^9, 1 \leq A_i \leq 10^7$$

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Constraints

$$1 \leq T \leq 102 \leq N \leq 1001 \leq K \leq \left\lfloor \frac{N}{2} \right\rfloor 0 \leq A_i < 2^7$$

Code snippets (also called starter code/boilerplate code)

This question has code snippets for C, CPP, Java, and Python.

Sample input

```
1
4
2
1 2 4 3
```

Sample output

```
4
```

Explanation

The first line contains the number of test cases, $T = 1$.

The first test case[< Previous Question](#)[Next Question >](#)

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Explanation

The first line contains the number of test cases, $T = 1$.

The first test case

Given

- $N = 4, K = 2$
- $A = [1, 2, 4, 3]$

Approach

- There is 1 subsequence of length 4.
- $B = [1, 2, 4, 3], F(B) = (1 \setminus \{1\}) \cup (2 \setminus \{1\}) \cup (4 \setminus \{1\}) \cup (3 \setminus \{1\}) = 4$.

Thus, the answer is 4.

Note:

Your code must be able to print the sample output from the provided sample input. However, your code is run against multiple hidden test cases. Therefore, your code must pass these hidden test cases to solve the problem statement.

Limits

Time Limit: 2.0 sec(s) for each input file

Memory Limit: 256 MB

Source Limit: 1024 KB

Scoring

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- $N = 4, K = 2$
- $A = [1, 2, 4, 3]$

Approach

- There is 1 subsequence of length 4.
- $B = [1, 2, 4, 3], F(B) = (1 \setminus \{\}) 2 \setminus \{\text{lo plus}\} (4 \setminus \{\}) 3 = 4.$

Thus, the answer is 4.

Note:

Your code must be able to print the sample output from the provided sample input. However, your code is run against multiple hidden test cases. Therefore, your code must pass these hidden test cases to solve the problem statement.

Limits

Time Limit: 2.0 sec(s) for each input file

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Score is assigned if any testcase passes

Allowed Languages

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00:57:21

902466566600@googl...

C++17 (g++ 10.3.0)

Auto-complete connection
failed.

Reconnect

```
1 #include<bits/stdc++.h>
2 using namespace std;
3
4 int solve (int N, int K, vector<int> A) {
5     // Write your code here
6
7 }
8
9 int main() {
10
11     ios::sync_with_stdio(0);
12     cin.tie(0);
13     int T;
14     cin >> T;
15     for(int t_i = 0; t_i < T; t_i++)
16     {
17         int N;
18         cin >> N;
19         int K;
20         cin >> K;
21         vector<int> A(N);
22         for(int i_A = 0; i_A < N; i_A++)
23         {
24             cin >> A[i_A];
25         }
26     }
```

View test cases ^

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+0.63%

ENG
IN



ail



YouTube



Maps

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Question 1



Max. score: 30.00

Maximum length subarray

You are given the following:

- Integers N , X , and Y
- Array A as $A_1, A_2, \dots, A_N$ of length N

You can perform the following two types of operation on array A :

1. *Type 1*: Choose an index i ($1 \leq i \leq N$), replace A_i by $A_i - 1$
2. *Type 2*: Choose an index i ($1 \leq i \leq N$), replace A_i by 0

Task

Determine the maximum length subarray of array A that can be obtained such that the value of all the elements in subarray is zero, after applying at most X and Y operations of *type 1* and *type 2*, respectively.

Notes

- The subarray of array A with length N is a contiguous part of

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Now



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after applying at most X and Y operations of *type 1* respectively.

Notes

- The subarray of array A with length N is a contiguous part of array A , which means that the subarray must be $A_l, A_{l+1}, \dots, A_r$ for $1 \leq l \leq r \leq N$.
- Assume 1 based indexing.

Example

Assumptions

- $N = 4$
- $X = 2$
- $Y = 1$
- Array $A = [4, 2, 0, 1]$

Approach

- Applying one operation of *type 2* on index 2, to replace A_2 by 0 and array A as $[4, 0, 0, 1]$
- Applying one operation of *type 1* on index 4, to replace A_4 by

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C



F1



F2



F3



F4



F5



F6



F7

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Function description

Complete the *LongestSubarray* function provided in the editor. This function takes the following 4 parameters and returns the maximum length subarray of array *A* that can be obtained such that the value of all the elements in subarray is zero, after applying **at most** *X* and *Y* operations of *type 1* and *type 2*, respectively:

- *N*: Represents the size of array *A*
- *X*: Represents the integer *X*
- *Y*: Represents the integer *Y*
- *A*: Represents the array *A* of length *N*

Input format

Note: This is the input format that you must use to provide custom input (available above the **Compile and Test** button).

- The first line contains an integer *T* denoting the number of test cases. *T* also denotes the number of times you have to run the *LongestSubarray* function on a different set of inputs.
- For each test case:
 - The first line contains a single integer *N*.

The second line contains a single integer *M*.

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Output format

For each test case in a new line, print the maximum length subarray that can be obtained such that the value of all the elements in the subarray is zero, after applying at most X and Y operations of *type 1* and *type 2* respectively.

Constraints

$$1 \leq T \leq 7$$

$$1 \leq N \leq 10^5$$

$$0 \leq A_i \leq 10^9$$

$$0 \leq X \leq 10^{14}$$

$$0 \leq Y \leq 10^9$$

Code snippets (also called starter code/boilerplate code)

This question has code snippets for C, CPP, Java, and Python.

Sample input

```
1
6
4
2
4 2 1 3 2 5
```

Sample output

```
4
```

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Now



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The first line represents the number of test cases, $T = 1$.

For test case 1

Given

- $N = 6$
- $X = 4$
- $Y = 2$
- Array $A = [4, 2, 1, 3, 2, 5]$

Approach

- Applying one operation of type 2 on index 1, to replace (A_1) by 0 and array A as $[0, 2, 1, 3, 2, 5]$.
- Applying one operation of type 2 on index 4, to replace (A_4) by 0 and array A as $[0, 2, 1, 0, 2, 5]$.
- Applying one operation of type 1 on index 2, to replace (A_2) by 1 and array A as $[0, 1, 1, 0, 2, 5]$.
- Applying one operation of type 1 on index 2, to replace (A_2) by 0 and array A as $[0, 0, 1, 0, 2, 5]$.
- Applying one operation of type 1 on index 3, to replace (A_3) by 0 and array A as $[0, 0, 0, 0, 2, 5]$.
- Therefore, the maximum length subarray of array A with all zeros is 4, after applying at most X and Y operations of type 1 and type 2, respectively.

Note:

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- Applying one operation of *type 1* on index 2, to replace $\backslash(A_2 \backslash)$ by 0 and array A as [0, 1, 1, 0, 2, 5].
- Applying one operation of *type 1* on index 2, to replace $\backslash(A_2 \backslash)$ by 0 and array A as [0, 0, 1, 0, 2, 5].
- Applying one operation of *type 1* on index 3, to replace $\backslash(A_3 \backslash)$ by 0 and array A as [0, 0, 0, 0, 2, 5].
- Therefore, the maximum length subarray of array A with all zeros is 4, after applying at most X and Y operations of *type 1* and *type 2*, respectively.

Note:

Your code must be able to print the sample output from the provided sample input. However, your code is run against multiple hidden test cases. Therefore, your code must pass these hidden test cases to solve the problem statement.

Limits

Time Limit: 5.0 sec(s) for each input file

Memory Limit: 256 MB

Source Limit: 1024 KB

Scoring

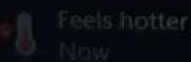
Score is assigned if any testcase passes

Allowed Languages

Bash, C, C++14, C++17, Clojure, C#, D, Erlang, F#, Go, Groovy, Haskell, Java 8, Java 14, JavaScript(Node.js), Julia, Kotlin, Lisp (SBCL), Lua, Objective-C, OCaml, Octave, Pascal, Perl, PHP, Python, Python 3, Python 3.8, R(RScript), Racket, Ruby, Rust, Scala, Swift, TypeScript, Visual Basic

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Question 2

Max. score: 30.00

Find palindromes

You are given a tree T with N nodes and the tree rooted at node 1. Every node has a character $C[i]$ assigned to it. You are given Q queries of the following format:

Query: u

You must find whether string S that is generated for node u is palindromic or not. String S for a node u is generated as follows:

```
S = Empty
makeString(u)
{
    For all v = Child of u
        makeString(v)

    S += C[u];
}

** Child of v should be traversed in order of increasing Node number
```

Input format

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Input format

- The first line contains an integer N .
- Next $N - 1$ lines contain two space-separated integers u and v denoting an edge between node u and node v .
- The next line contains N space separated characters denoting $C[i]$ for all i from 1 to N .
- The next line contains an integer Q .
- Next Q lines contain an integer u denoting the node for the query.

Output format

Print Q lines and in each line, print **1** if S is palindromic. Otherwise, print **0**.

Note: Use fast I/O

Constraints $1 \leq N, Q \leq 200000$ $C[i]$ contains lowercase Latin Letters.

Sample input



Sample output



5

0

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Note: Use fast I/O

Constraints $1 \leq N, Q \leq 200000$ $C[i]$ contains lowercase Latin Letters.

Sample input



Sample output



```
5
1 2
1 3
2 4
2 5
a b a b c
2
1
2
```

```
0
1
```

ExplanationFor node 1, $S = "bcbaa"$ For node 2, $S = "bcb"$ **Note:**

Your code must be able to print the sample output from the provided sample input. However, your code is run against multiple hidden test cases. Therefore, your code must pass these hidden test cases to solve the problem statement.

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Explanation

For node 1, S = "bcbaa"

For node 2, S = "bcb"

Note:

Your code must be able to print the sample output from the provided sample input. However, your code is run against multiple hidden test cases. Therefore, your code must pass these hidden test cases to solve the problem statement.

Limits

Time Limit: 2.0 sec(s) for each input file

Memory Limit: 512 MB

Source Limit: 1024 KB

Scoring

Score is assigned if any testcase passes

Allowed Languages

Bash, C, C++14, C++17, Clojure, C#, D, Erlang, F#, Go, Groovy, Haskell, Java 8, JavaScript(Node.js), Julia, Kotlin, Lisp (SBCL), Lua, Objective-C, OCaml, Octave, Pascal, Perl, PHP, Python, Python 3, R(RScript), Racket, Ruby, Rust, Scala, Swift, TypeScript, Visual Basic

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Upcoming
Earnings



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1

String machine

2

You are given a string $S1$ of length N .

You have a machine that will generate a string Q of length M using a string $S2$ of length M .

However, the machine will not generate the string Q at once. Initially, you're given a string Q filled with the character " ". The machine will add the characters of string $S2$ at specific positions one by one in string Q each second. You also have a permutation P of length M .

At the i^{th} second of time, the machine will add $P[i]^{th}$ character of $S2$ to $P[i]^{th}$ position of Q . So, eventually at the end of M^{th} second, the whole string $S2$ will be ready as a string Q .

You want to stop the machine as soon as possible (even if Q is not fully made) and the string Q should have $S1$ as a subsequence at that moment when the machine is stopped.

Notes

- A subsequence is a sequence that can be derived from another sequence by deleting some or no elements without changing the order of the remaining elements.

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Notes

- A subsequence is a sequence that can be derived from another sequence by deleting some or no elements without changing the order of the remaining elements.
- A permutation of length X is a sequence of integers from 1 to X containing each number exactly once.
- 1-based indexing is used.

Task

Determine what is earliest time (in seconds) after which the machine can be stopped such that the string Q has $S1$ as a subsequence. Print -1 if it is not possible.

Example

Assumptions

- $N = 3$
- $S1 = "abc"$
- $M = 5$
- $S2 = "adbec"$

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```
3 #include
4
5 int solve
6     // Wr
7
8 }
9
10 int main
11     int
12     scan
13     for(
14 {
15
16
17
18
19
20
21
22
23
24
25
26
```



Determine what is earliest time (in seconds) after which the machine can be stopped such that the string Q has $S1$ as a subsequence. Print -1 if it is not possible.

Example

Assumptions

- $N = 3$
- $S1 = "abc"$
- $M = 5$
- $S2 = "adbec"$
- $P = \{1, 4, 3, 5, 2\}$

Approach

String Q is initially " $* * * * *$ ".

- $P[1] = 1$, after first second, 1st character is added, $Q = "a*****"$
- $P[2] = 4$, after second second, 4th character is added, $Q = "a**e**"$
- $P[3] = 3$, after third second, 3rd character is added, $Q = "a*be**"$
- $P[4] = 5$, after fourth second, 5th character is added, $Q = "a*bec**"$
- $P[5] = 2$, after fifth second, 2nd character is added, $Q = "abec**"$

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Approach

String Q is initially `"*****"`.

- $P[1] = 1$, after first second, 1st character is added, $Q = "a*****"$
- $P[2] = 4$, after second second, 4th character is added, $Q = "a**e*"$
- $P[3] = 3$, after third second, 3rd character is added, $Q = "a*be*"$
- $P[4] = 5$, after fourth second, 5th character is added, $Q = "a*bec"$
- $P[5] = 2$, after fifth second, 2nd character is added, $Q = "adbec"$
- After the fourth second, you have $S1$ as a subsequence of string Q for the first time.

Hence, the answer is 4.

Function description

Complete the `solve` function provided in the editor. The function takes the following 5 parameters and returns the final answer:

- N : Represents the length of $S1$
- $S1$: Represents the string $S1$
- M : Represents the length of $S2$

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Successful

Question 1

Max. score: 30.00

Palindromic triplets

You are given a string S of length n . The string consists of lowercase English alphabets.

A triplet of non-overlapping pairs of indices $P1(i_1, j_1)$, $P2(i_2, j_2)$, $P3(i_3, j_3)$ ($i_1 \leq j_1 < i_2 \leq j_2 < i_3 \leq j_3$) is said to be *good* if substrings formed from these pairs are palindrome.

Formally, $S[i_1..j_1]$, $S[i_2..j_2]$ and $S[i_3..j_3]$ are palindromes.

Task

Determine the number of good triplets.

Notes

- 0 based indexing is followed.
- A *substring* is the sequence of consecutive elements of the string. For example, in string "abc" the substrings are as follows:
 - "" - empty string

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Explanation

The first line represents the number of test cases, $T = 1$.

The first test case

Given

- $n = 5$
- $S = \text{"efcdc"}$

Approach

Triples of non-overlapping pairs of indices which are palindrome are:

- $P1(0, 0) = \text{"e"}, P2(1, 1) = \text{"f"}, P3(2, 4) = \text{"cdc"}$.
- All triplets with substring length of 1 are:
 - $P1(0, 0), P2(1, 1), P3(2, 2)$.
 - $P1(0, 0), P2(1, 1), P3(3, 3)$.
 - $P1(0, 0), P2(1, 1), P3(4, 4)$.
 - $P1(0, 0), P2(2, 2), P3(3, 3)$.
 - $P1(0, 0), P2(2, 2), P3(4, 4)$.
 - $P1(0, 0), P2(3, 3), P3(4, 4)$.
 - $P1(1, 1), P2(2, 2), P3(3, 3)$.
 - $P1(1, 1), P2(2, 2), P3(4, 4)$.
 - $P1(1, 1), P2(3, 3), P3(4, 4)$.
 - $P1(2, 2), P2(3, 3), P3(4, 4)$.

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- P1 (0, 0), P2 (2, 2), P3 (4, 4)
- P1 (0, 0), P2 (3, 3), P3 (4, 4)
- P1 (1, 1), P2 (2, 2), P3 (3, 3)
- P1 (1, 1), P2 (2, 2), P3 (4, 4)
- P1 (1, 1), P2 (3, 3), P3 (4, 4)
- P1 (2, 2), P2 (3, 3), P3 (4, 4)

Hence, the answer is 11.

Note:

Your code must be able to print the sample output from the provided sample input. However, your code is run against multiple hidden test cases. Therefore, your code must pass these hidden test cases to solve the problem statement.

Limits

Time Limit: 2.0 sec(s) for each input file

Memory Limit: 256 MB

Source Limit: 1024 KB

Scoring

Score is assigned if any testcase passes

Allowed Languages

Bash, C, C++14, C++17, Clojure, C#, D, Erlang, F#, Go, Groovy, Haskell, Java 8, Java 14, Java 17, JavaScript(Node.js), Julia, Kotlin, Lisp (SBCL), Lua, Objective-C, OCaml, Octave, Pascal, Perl, PHP, Python, Python 3, Python 3.8, R(RScript), Racket, Ruby, Rust, Scala, Swift, TypeScript, Visual Basic

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Question 2

Max. score: 30.00

Prime tree

You are given an undirected tree T consisting of N nodes rooted at node 1. You have to assign value to each node of the tree such that:

- The assigned value must be a prime number less than or equal to 10^2 .
- The Sum of values assigned to any pair of adjacent nodes is not a prime number.

Task

Determine the total number of valid assignments. Since, this number can be very large output it modulo $10^9 + 7$.

Example

Assumptions

- $T = 1$
- $N = 2$
- $edges = \{(1, 2)\}$

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edges = {(1,2)}

Approach

- You cannot assign a given pair of values to node 1 and node 2, as their sum is a prime number.
 - (2,3), (2,5), (2,11), (2,17), (2,29), (2,41), (2,59), (2,71)
 - (3,2), (5,2), (11,2), (17,2), (29,2), (41,2), (59,2), (71,2)

Hence, total number of valid assignments is $625 - 16 = 609$. The answer is 609.

Function description

Complete the `validAssignment` function provided in the editor. This function takes the following 2 parameters and returns the total number of valid assignments:

- N*: Represents the number of nodes in the tree
- edges*: Represents the edges of the tree

Input format

Note: This is the input format that you must use to provide custom input (available above the **Compile and Test** button).

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input (available above the **Compile and Test** button).

- The first line contains a single integer T , which denotes the number of test cases. T also specifies the number of times you have to run the `validAssignment` function on a different set of inputs.
- For each test case:
 - The first line contains an integer N .
 - The next $N - 1$ line contains 2 space-separated integers denoting an edge.

Output format

For each test case, print the total number of valid assignments modulo $10^9 + 7$ in a new line.

Constraints

$$1 \leq T \leq 10$$

$$1 \leq N \leq 10^4$$

$$1 \leq u, v \leq N$$

Code snippets (also called starter code/boilerplate code)

This question has code snippets for C, CPP, Java, and Python.

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Haze



Search





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Question 1

Successfully signed in

Subtree XOR

You are given the following:

- A tree with N nodes
- An array A denoting the value of each node
- A non-negative integer K

You can do the following operation on the tree *at most once*:

- Pick a node r and make it the root of the tree
- In the new tree, pick a node x , and for each of the node in the subtree in x update $A_{subtree} = A_{subtree} \oplus K$

Task

Determine the maximum sum of values of all the nodes in the tree that can be obtained.

Notes

- 1- Based indexing.

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Mostly cloudy



Search



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Task

- 1 Determine the maximum sum of values of all the nodes in the tree that can be obtained.
- 2

Notes

- 1- Based indexing.
- $\oplus$ denotes Bitwise-xor operator.
- A subtree of a node contains all the nodes that are descendants of that node and the node itself.

Example

Assumptions

- $N = 3$
- $edges = [(1, 2), (1, 3)]$
- $A = [1, 1, 3]$
- $K = 3$

Approach

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Task

- 1 Determine the maximum sum of values of all the nodes in the tree that can be obtained.
- 2

Notes

- 1- Based indexing.
- $\oplus$ denotes Bitwise-xor operator.
- A subtree of a node contains all the nodes that are descendants of that node and the node itself.

Example

Assumptions

- $N = 3$
- $edges = [(1, 2), (1, 3)]$
- $A = [1, 1, 3]$
- $K = 3$

Approach

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Approach

You perform the following operation to get the maximum sum:

- Pick $x = 3$ as the root of the tree
- Update the value in subtree of $x = 1$
- The new array becomes $A = [2, 2, 3]$

The sum obtained is $2 + 2 + 3 = 7$ which is maximum. Hence, the answer is 7.

Function description

Complete the function `maximumSum` provided in the editor. This function takes the following 4 parameters and returns the maximum sum that can be obtained:

- N : Represents an integer denoting the number of nodes in the tree
- `edges`: Represents a 2-D array of size $(N - 1) \times 2$ denoting the edges in the given tree
- A : Represents an array denoting the value of each of the node
- K : Represents an integer denoting the value used in the operation

Input format

State this is the input format that you must use to read the input from the user.

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